

Jie Bao

PERSONAL INFORMATION

WEBSITE / EMAIL / CITY: jiebao.ca / jiebao995@gmail.com / Montreal, Quebec, Canada

COMPUTER SKILLS

PYTHON, SPARK, DATABRICKS, FASTAPI, K8S, SQLAlchemy, GIT, DOCKER, BLENDER

WORK EXPERIENCE

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| 2025-Now | Research Intern , SUPERVISED BY PROFESSOR ADRIEN GRUSON AND PIERRE POULIN
Designed and ran experiments for monocular 3D face reconstruction using Mitsuba 3 and the FLAME morphable model, building optimization pipelines that jointly refine geometry, lighting, and appearance.

Improved training stability and accuracy by using SAM to isolate valid pixels and reduce background ambiguity. Implemented a depth-based regularization for geometric consistency. Trained a VAE-GAN for HDR environment-map estimation, providing a learned prior that reduces ambiguity between surface reflectance and illumination |
| 2025 | Data Scientist , MOOV AI A PUBLICIS GROUPE COMPANY
Developed machine-learning systems for clients including Énergir and CSA . Delivered a data driven forecasting platform cutting critical-threshold errors to 9.3% at 124-hour lead time.

Co-developed Emilia, an AI correction assistant with French teachers at CSA, redesigning key pipeline components and reducing error rates by 50%. Worked closely with client stakeholders to refine requirements and validate deployed solutions. |
| 2022-2025 | Data Engineering Specialist , CAE
Building internal Data Platform - CAE Data Catalog: Implemented robust data quality and observability backends using event-driven, autoscaling data pipelines with Spark. |
| 2019-2021 | Graduate Student Researcher , COMPUTATIONAL AEROSPACE LAB
Created an end-to-end ML pipeline for turbulent production values. Achieved over 90% R^2 accuracy. Practiced public speaking as a researcher and gave bi-weekly presentations as teaching assistant - AERO 490 . |

EDUCATION

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| CURRENT | Mila - Université de Montréal , Montreal, Quebec, Canada
COMPUTER SCIENCE MASTER STUDENT (PART-TIME)

<i>Realistic Image Synthesis - ECSE 546: A</i>
<i>Applied Machine Learning - COMP 551: A</i>
<i>Natural Language Processing with Deep Learning - IFT 6289: A-</i> |
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LANGUAGES

FRENCH, ENGLISH, CHINESE (MANDARIN): Fluent